

Chapter 14: More Complexity Ideas

14.1 Approximation Algorithms

An *approximation algorithm* is one where the answer is guaranteed to be “close” to the actual. For example, consider vertex cover. Here is a factor-2 approximation algorithm, meaning it produces a vertex cover that is guaranteed to be at most twice the optimal size:

While there exists some edge, take both ends of the edge and remove.

Note that for every edge we remove, we need at least one end in every vertex cover.

It is known that: if $P \neq NP$, then there is no polynomial-time algorithm to approximate domination number within a constant factor. The best known algorithm has a logarithmic factor. Similarly, there exists a constant $\epsilon > 0$ such that one cannot approximate chromatic number to within a factor of n^ϵ .

For a long time, the best approximation result for 3-colorable graphs was: that one could find a coloring using $O(\sqrt{n})$ colors:

While there exists a vertex degree at least $\sqrt{n}$, 2-color its neighbors and remove them. Thereafter greedily color with $\sqrt{n} + 1$ colors.

14.2 Polynomial Hierarchy

One can also define NP as follows: a problem B is in NP if there is a polynomial-time decidable problem A of ordered pairs $\langle x, y \rangle$ such that

$$B = \{ x : \exists y : \langle x, y \rangle \in A \}.$$

The *polynomial hierarchy* allows for multiple quantifiers. For example, co-NP is the complement of sets in NP, or equivalently, we replace $\exists$ by $\forall$ in the above (and A by its complement).

The class Σ_2^P is the collection of all problems one gets from a polynomial-time decidable A of triples by defining

$$B = \{ x : \exists y \forall z : \langle x, y, z \rangle \in A \}.$$

(The P is for “polynomial-time”.) For example, a graph is said to be **2-clique colorable** if one can color the vertices with 2 colors such that no *maximal* clique is monochromatic. This problem is in Σ_2^P . (Indeed this problem is Σ_2^P complete.) The complement of Σ_2^P is Π_2^P . In this notation, NP is Σ_1^P . One can then define further classes by adding more (alternating) quantifiers. The union of these is called the *polynomial hierarchy*.

14.3 PSPACE

The set PSPACE is the set of all decision problems that can be decided in polynomial space. It can be shown that it includes all of the polynomial hierarchy. Graph-theoretic games are often PSPACE-complete. These are games where two players take turns, where the goal of one player is to create something and the goal of the other player is to avoid something. For example, a game where k colors are given and the vertices are numbered, and the player must properly color the next vertex.

14.4 #P

The class “sharp-P” or “number-P” is a set of counting problems. Surprisingly, counting the number of perfect matchings in a graph has the same complexity as counting the number of 3-colorings (both are #P-complete), even though the existence questions are not (or rather are believed not) the same complexity.

14.5 Fixed-parameter tractability

Some NP-complete problems have a parameter as part of the input. Sometimes fixing the parameter does not change the complexity. For example, the chromatic

number problem is NP-complete; and even if one restricts to asking about 3-colorability, the problem remains so. In contrast, the genus-of-a-graph problem is NP-complete. But for any fixed genus there is a fast algorithm that determines whether a graph embeds on that surface.

A parameter π is called ***fixed-parameter tractable*** if there is an algorithm for the decision problem $\pi(G) \leq k$ that runs in time $f(k)n^C$ for some constant C and function f . For example, the genus is such a parameter. On the other hand domination is believed not to be so.

Vertex cover is fixed-parameter tractable. Here is the algorithm. It uses ***kernelization***; meaning we trim the graph to a constant-size graph. To determine if the vertex cover number is at most k :

- (1) *If there is a vertex of degree more than k , then take v . (If we don't take v , then we need more than k vertices.) Repeat.*
- (2) *Throw away isolated vertices.*
- (3) *If the graph at this point (meaning $\Delta \leq k$) has more than k^2 edges, then it does not have a vertex cover of size k .*
- (4) *If not, we have a graph with at most k^2 edges and no isolates. It follows that the number of vertices is at most $2k^2$. And now, simply use exhaustive search. This takes time exponential in k but does not depend on the order of the original graph.*